

Engineering interview exercise

Workflow evaluator

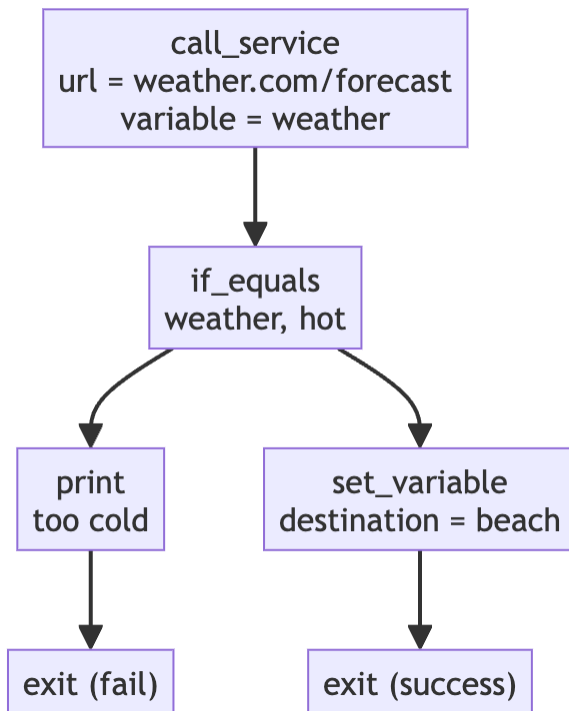
We want to build a system that receives and runs workflows.

The supported actions in a workflow are:

- `set_variable`: set a variable value
- `call_service`: calls an external web service and stores the result into a variable
- `read_file`: reads a file into a variable
- `print`: prints a variable or text
- `if_equals`: check if a variable is equal to a value or another variable
- `if_file_exists`: performs different actions based on whether a file exists
- `exit`: exit the workflow with either success or failure

The system takes in a workflow (see format below), and returns success/failure (based on the `exit` action), the workflow output (generated by the `print` action), and all variables with their values.

Example workflow



Format

Specifying the format of the workflow is in your responsibility, as part of this exercise.

Assume you have a brilliant frontend guy who can design an amazing graphical editor. The user will drag-and-drop actions, connect them with arrows, etc. The frontend app will convert the workflow into a string in the format you'll specify, and that string will be the input to your workflow evaluator.

Language and framework

Use whatever is most convenient to you, as long as it's not exoteric or trivializes the exercise.

In other words, don't use COBOL or `import { runner } from "workflow-runner"`. Anything else is fine.

Scope of the exercise

Since this is more of a design exercise than a coding exercise, we don't expect working, or even compiling, code. feel free to write comments instead of trivial stuff -

JavaScript

```
function readFile(path): string {  
    // TODO: read file and return content  
}
```

We do, however, expect a robust design. create a solid foundation and describe what you'll do to make it production-worthy.